

Blind Image Restoration via Dual-Conditioned Latent Diffusion

Master's Thesis Defense

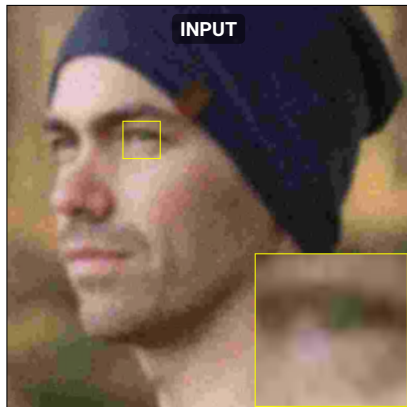
Carmelo Calzerano

Department of Information Engineering
University of Pisa

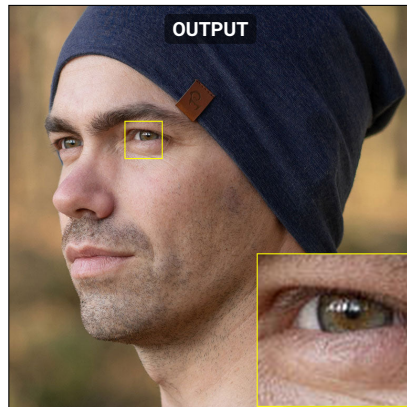
Supervisor: Prof. Tiberio Uricchio

2026

What is Blind Image Restoration?



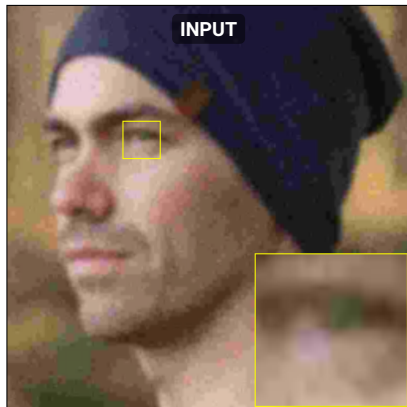
Degraded Image



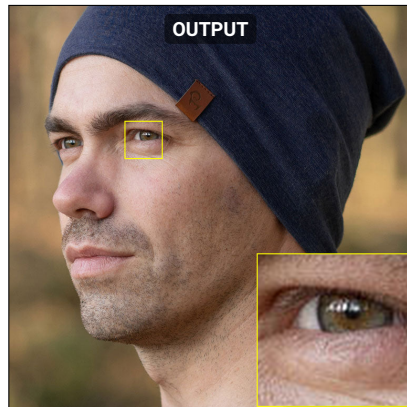
Clean Image

- ▷ Recover the clean image when the **degradation is unknown**

What is Blind Image Restoration?



Degraded Image



Clean Image

Blind Restoration:

No information about the degradation is available at inference time

What Does Degradation Look Like?

Common sources of degradation:

Blur

camera shake, lens defocus, motion

Noise

sensor electronics, low light

Compression

JPEG artifacts, streaming

Resolution loss

downsampling, digital zoom

All combine **simultaneously** and **unpredictably**

Part of the information is **permanently lost**

→ multiple valid reconstructions exist

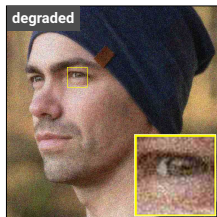


The model must choose

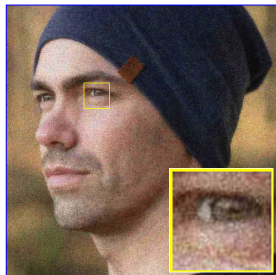
among plausible solutions



Three Approaches to Restoration



REGRESSION



HIGH FIDELITY

(but low perceptual quality)

Samples
a plausible solution
may hallucinate details

Averages
across solutions
tends to over-smooth details

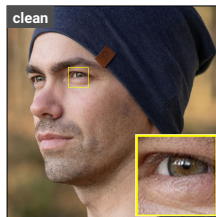
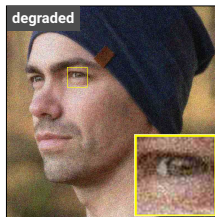
GENERATIVE



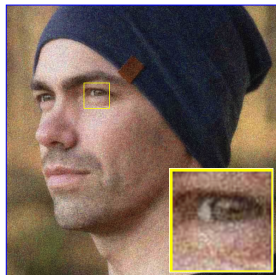
HIGH QUALITY

(but low pixel fidelity)

Three Approaches to Restoration



REGRESSION



HIGH FIDELITY

(but low perceptual quality)



HIGH FIDELITY + HIGH QUALITY

GENERATIVE



HIGH QUALITY

(but low pixel fidelity)

Research Question

Can we achieve **both** the **pixel fidelity** of regression models
and the **perceptual quality** of generative models?



Our approach:

Repurpose a pretrained generative model,
explicitly **specializing** it for the restoration task

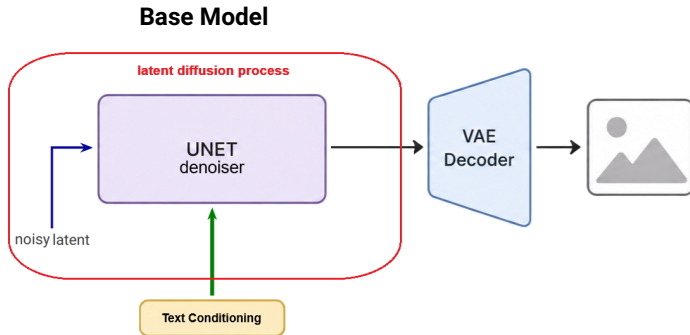
Requirements

- **Dense** input conditioning
- Pixel-level **accuracy**
- Preserve **generative prior**

Design Challenges

- **Catastrophic forgetting**
- Loss of **fine details** (compression bottleneck)
- **Hallucination** of plausible content

Repurposing Stable Diffusion for Restoration



Stable Diffusion (Rombach et al., 2022)

Text-to-image generation via iterative denoising in latent space

Rich prior

large-scale pretraining

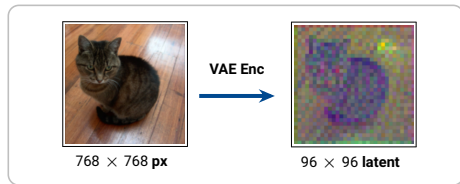
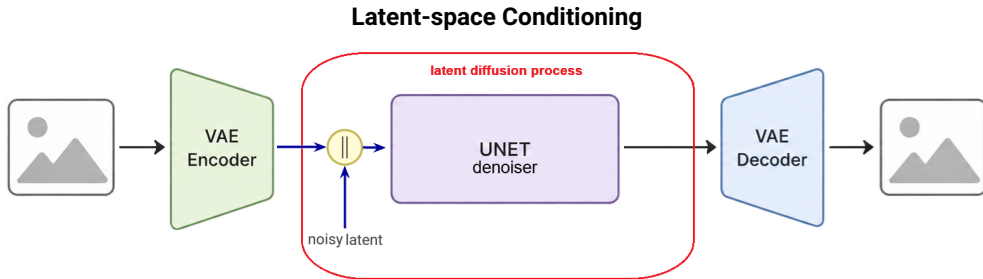
Efficient

8 × latent compression

Adaptable

Marigold, ControlNet

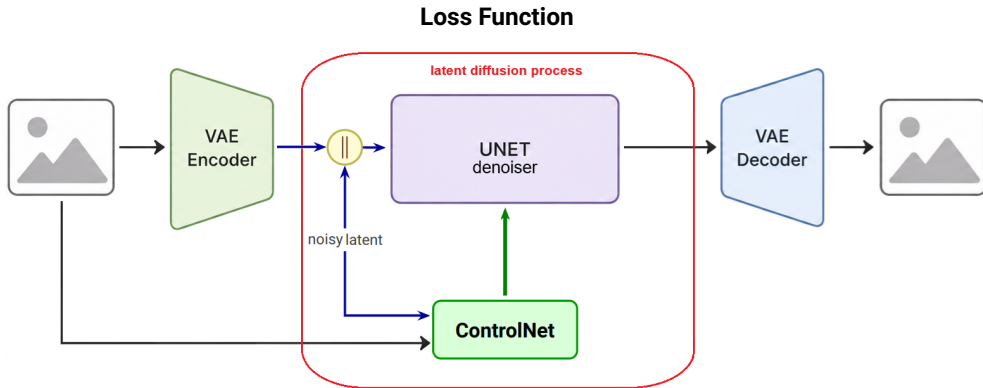
Repurposing Stable Diffusion for Restoration



dense structure/color signal

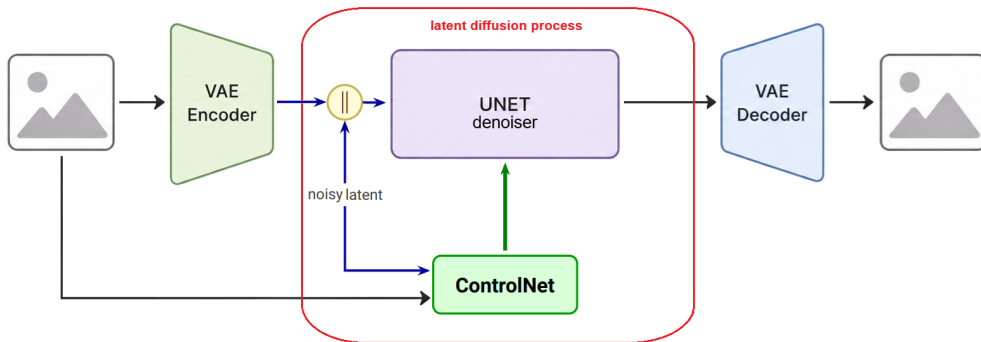
$8\times$ reduced resolution

Repurposing Stable Diffusion for Restoration



$$\mathcal{L} = \underbrace{\mathcal{L}^{\text{lat}}}_{\text{latent MSE}} + \lambda_{\text{LPIPS}} \underbrace{\mathcal{L}^{\text{LPIPS}}}_{\text{perceptual}} + \lambda_{\text{pixel}} \underbrace{\mathcal{L}^{\text{pixel}}}_{\text{pixel L1}}$$

Complete Model Architecture



Differential learning rates (for UNet/CNet) →

ControlNet (pixel-space pathway) →

Dense dual conditioning (8ch + CN) →

avoids forgetting

bypasses compression bottleneck

mitigates hallucination

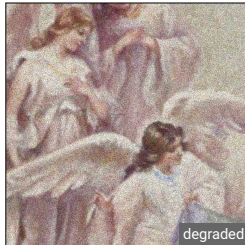
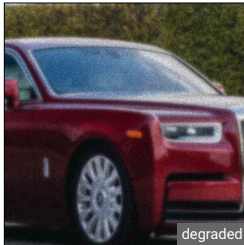
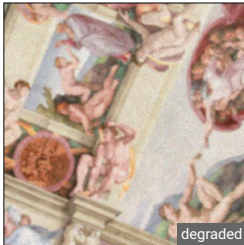
Training Setup

Training Dataset:

- LAION-Aesthetics subset
- $\sim 20,000$ images
- 768×768 random crops

Synthetic Degradation:

- Real-ESRGAN pipeline
blur $\rightarrow$ resize $\rightarrow$ noise $\rightarrow$ JPEG



15k iterations | batch size 16 | ~ 20 h on NVIDIA L40S

Evaluation Overview

Two evaluation axes:

Synthetic (DIV2K)

Same degradation as training
→ Can the model **learn**?

Real (RealSR Canon/Nikon)

Unseen optical degradation
→ Can the model **generalize**?

Two metric families:

Fidelity

closeness to ground truth

PSNR pixel accuracy (dB)

SSIM structural similarity

ΔE color difference

Perceptual quality

naturalness

LPIPS learned perceptual distance

Comparison models:

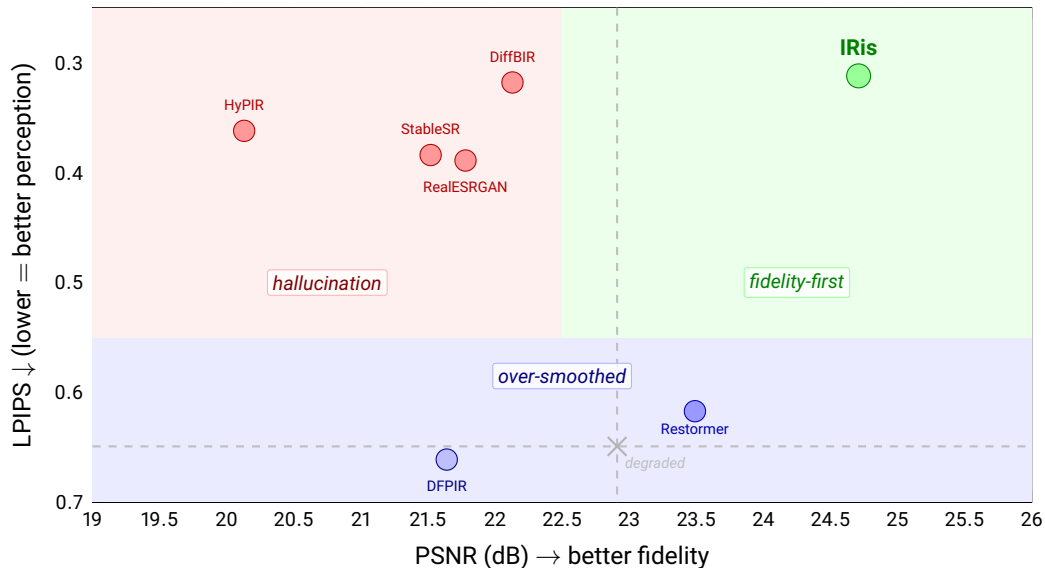
Real-ESRGAN (GAN, 2021) | **Restormer** (Regression, 2022) | **DiffBIR** (Two-stage Diff., 2024) | **HyPIR** (GAN-distilled, 2025)

Evaluation Results

	Synthetic (DIV2K)				Real (RealSR L4)			
	PSNR↑	SSIM↑	ΔE ↓	LPIPS↓	PSNR↑	SSIM↑	ΔE ↓	LPIPS↓
IRis	24.71	0.670	4.54	0.312	26.04	0.766	4.05	0.282
Restormer	<u>23.49</u>	0.543	5.70	0.617	<u>25.91</u>	<u>0.742</u>	<u>4.16</u>	0.452
Real-ESRGAN	21.78	0.563	6.18	0.389	23.24	0.698	5.12	0.387
DiffBIR	22.13	<u>0.568</u>	<u>5.56</u>	<u>0.318</u>	23.00	0.643	5.05	0.354
HyPIR	20.13	0.498	7.26	0.362	21.00	0.635	6.48	<u>0.320</u>
Degraded	22.91	0.474	6.36	0.649	25.98	0.743	4.21	0.428

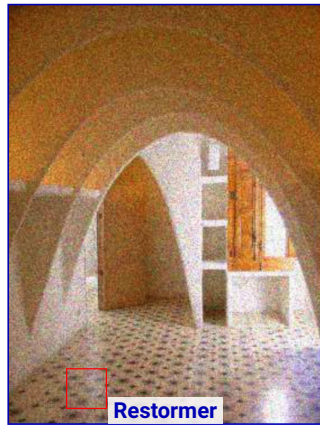
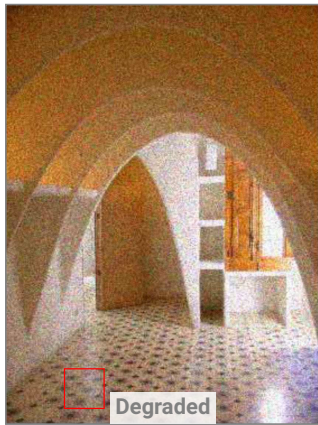
IRis ranks **1st on all metrics**
on both **synthetic** and **real** degradation

Fidelity vs. Perceptual Quality



Visual Comparison

The visual results are consistent with the quantitative evaluation



Regression preserves structure but **loses detail**

Visual Comparison

The visual results are consistent with the quantitative evaluation



Both generative models produce **sharp** output

Visual Comparison

The visual results are consistent with the quantitative evaluation



On close inspection, DiffBIR **hallucinates**; IRis is **faithful**

Research question:

Can we achieve **both** the **pixel fidelity** of regression models
and the **perceptual quality** of generative models?

The experimental evidence suggests that the answer is

YES

Repurposing a pretrained diffusion model for restoration can
improve the fidelity–quality tradeoff
(at least under the experimental conditions considered)

Key design choices: Dual conditioning | Joint fine-tuning | Pixel-space perceptual loss

Limitations: training scale, degradation diversity | **Future work:** single-step distillation, broader generalization

Thank you

Questions?